

Food Process Technology

Science

May, 2026



Food Process Technology (A06319)

Short Title: Food Process Technology
Faculty Science and Computing
Department Science
Cluster Food Science
Author EDUGGAN
Credits 5

Level: Introductory

Description of Module / Aims

This module introduces students to food processing and preservation methods currently used in the food industry. It focuses on both the theory of the methodology and the equipment used for conventional processing methods.

Programmes

SCIE-0033	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 4 / M
SCIE-0033	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	2 / 4 / M
SCIE-0033	BSc in Food Science with Business (WD_SFDSC_D)	2 / 4 / M

Indicative Content

- Thermal processing: thermisation, pasteurisation, sterilisation
- Concentration and dehydration: evaporation, spraydrying, fluidised bed, freeze-drying
- Cooling and freezing of foods
- Size reduction:- milling, mincing, grinding
- Fermentation:- cheese and yogurt
- Processing of meat in abattoir, meat products
- Packaging: types and purpose

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe typical food process flow diagrams used in the food industry.
2. Outline the principles of unit processes for the food industry.
3. Compare and contrast items of equipment for different unit processes.
4. Compare different preservation and packaging technologies for fresh and manufactured foods with enhanced shelf-life.
5. Complete a range of food process operations.
6. Report on a range of food process operations.

Learning and Teaching Methods

- Lectures will provide the theory and design of equipment used for food production and preservation. Case studies will be used to elucidate unit processes.
- Laboratory practicals will give students experience in performing food processing experiments, analysing data and writing reports.
- Taste panels will be used to introduce students to sensory analysis.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4
Continuous Assessment	50%	
<i>Practical</i>	50%	5,6

Assessment Criteria

- <40%:** Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with regards to use of associated laboratory skills, techniques and instrumentation.
- 40%-49%:** Demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regard to the use of associated laboratory skills, techniques and instrumentation.
- 50%-59%:** Demonstrate satisfactory understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.
- 60%-69%:** Demonstrate sound knowledge of the all topics contained in the learning outcomes and display good competence with regards to use of associated laboratory instrumentation and very good laboratory practice.
- 70%-100%:** Demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to use of associated laboratory techniques and instrumentation and excellent laboratory practice.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Essential Material

- Brennan, J.G. and A.S. Grandison. *_Food Processing Handbook_*. 2nd ed.. Weinheim: Wiley-VCH, 2012.
- Earle, R.L. *_Unit Operations in Food Processing_*. Web Edition. New Zealand: NZIFST, 2004.
- Warriss, P.D. *_Meat Science:An Introductory Text_*. Wallingford, UK; New York: CABI Publishing, 2000.

Supplementary Material

- Lean, M.E.J. *_Fox and Cameron's Food Science, Nutrition & Health_*. 7th ed.. London: CRC Press, 2006.

Requested Resources

- Room Type: Chemistry Lab